

ABHINAV KAUSHIK

Principal Machine Learning Scientist, Frontier Research, AI for Drug Discovery

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SUMMARY

Senior AI/ML Engineer with 8+ years building retrieval, ranking, recommendation and production ML systems across Healthcare, Telecom and AIOps. Works with cross-encoders, embeddings, semantic retrieval, Python, SQL and PySpark, alongside production APIs and ML evaluation. Delivers measurable outcomes with stakeholders. Alongside this, I am building ThinkTree.AI, a non-profit AI learning platform used by NGO teachers to support students with ADHD.

PROFESSIONAL EXPERIENCE

Senior AI Engineer | United Health Group | Bengaluru, India

Nov 2025 – Present

- Fine-tuned a cross-encoder on positive and negative pairs to re-rank dense retrieval over AARP survey data, improving retrieval accuracy.
- Developed a RAG-based Project Discovery Assistant in LangGraph combining semantic retrieval with LLM reasoning, reaching a 90% match rate.
- Implemented a reranker for an existing recommendation system serving individual and child plan recommendations, increasing conversion by 2%.
- Built SQL and PySpark data pipelines automating medical-audit reporting on large healthcare datasets, cutting manual reporting effort by 95%.

Senior Machine Learning Engineer | Viavi Solutions | Chennai, India

Sep 2024 – Nov 2025

- Developed an internal RAG-based Project Discovery Assistant in LangGraph combining semantic retrieval with LLM reasoning, reaching a 90% match rate against existing organizational solutions.
- Built an event-correlation engine clustering alarms with Sentence-BERT and DBSCAN to trace the cascading effect of network alarm floods back to the root event, improving semantic alarm classification by 73.5%.
- Prototyped 15+ AI applications for AIOps and GeoPerformance clients from client requirements, with 7 moving into production or paid client work.

Data Scientist | Viavi Solutions | Bengaluru, India

Nov 2021 – Sep 2024

- Deployed a production SLA breach prediction system with LightGBM, FastAPI and MLflow, using GPT-3.5 for real-time root-cause analysis and remediation recommendations per breach, reducing downtime by 67%.
- Shipped anomaly detection and demand forecasting on 15+ KPIs across 500+ telecom network devices using Isolation Forest and Holt-Winters, enabling earlier detection and proactive network planning.
- Delivered technical POCs and consulting for 15+ clients across 3+ AIOps and ML platforms, contributing to an 11% increase in POC and mentoring revenue.

Data Analyst | Optum | Gurugram, India

Jun 2021 – Nov 2021

- Implemented a reranker for an existing recommendation system serving individual and child plan recommendations, optimizing renewal suggestions and increasing conversion by 2%.
- Developed a Python and Flask parser that automated extraction and transformation of raw data into structured datasets, feeding downstream analytics and reporting.

Data Engineer | Nippon Data Systems | Gurugram, India

Nov 2019 – May 2021

- Engineered SQL and PySpark data pipelines automating BAU medical-audit reporting on large healthcare datasets, cutting manual reporting effort by 95%.
- Wrote complex SQL queries with multi-table joins, CTEs and window functions, delivering ad-hoc reports and analyses for multiple stakeholder teams.
- Built real-time log observability for the COVID QRT team with Elasticsearch and Kibana, improving traceability efficiency by 80%.

Associate Consultant | Saksoft | Gurugram, India

Dec 2017 – Nov 2019

- Developed an XGBoost propensity model predicting personal-loan applications within 7 days of contact, enabling targeted outreach and customer prioritization.
- Built customer dashboards for financial and product insights with QlikSense, SQL and SAP HANA, increasing user engagement by 40%.

PROJECTS

Think Tree – ML Study Platform

- Built a non-profit AI-powered ML study platform for people with ADHD and context-switching challenges, designed to help users organize learning and maintain study context.

RFP Multi-Agent Proposal Automation Platform

- Built a multi-agent RFP platform where an LLM supervisor orchestrates intent detection, retrieval, proposal generation, document matching and market research, with RAG over historical proposals.

Financial Fraud Detection – Mar 2025

- Built a fraud detection and reasoning workflow on the JPMorgan Synthetic Dataset, pairing LightGBM scoring with LLM-generated explanations for interpretable financial-risk analysis.

Restaurant Feedback Analyzer – Aspect-Based Sentiment Analysis

- Built an NLP pipeline to extract review aspects and determine sentiment at the aspect level, enabling granular insights into customer experience.

EDUCATION

Master of Science, Machine Learning and AI Liverpool John Moores University	Jan 2021 – Jan 2023
Bachelor of Technology, Information Technology Jamia Hamdard	Aug 2013 – Aug 2017

TECHNICAL SKILLS

AI / GenAI: Generative AI, LLM, RAG, Agentic RAG, Agentic AI, LangGraph, LangChain, Claude, OpenAI, Embeddings, Cross-Encoder, Fine-Tuning, Hugging Face Transformers, NLP, Tool Orchestration, Amazon Textract

Machine Learning: PyTorch, Scikit-learn, XGBoost, LightGBM, Random Forest, Isolation Forest, LSTM, SBERT, DBSCAN, Time Series Forecasting, Demand Forecasting, Anomaly Detection, Statistical Modeling, Regression, Deep Learning

Evals / Observability: LLM Evals, LLM Evaluators, Agent Tracing, Observability, Elasticsearch, Kibana, Log Analysis

Programming / Data: Python, SQL, PySpark, Pandas, NumPy, FastAPI, Flask, Hive, Spark SQL, REST API, Pydantic, Unix

MLOps / Cloud: MLflow, Docker, Kubernetes, Airflow, DVC, Terraform, Jenkins, Bamboo, Bitbucket, GCP, Azure, Vercel

Data Platforms: Databricks, Databricks SQL, Apache Spark, Supabase

Databases: TimescaleDB, MySQL, Vector DB, pgvector, ArangoDB, Cosmos DB

Search: Apache Solr, Lucene Indexing, Information Retrieval

BI: Power BI, QlikSense, SAP HANA

LANGUAGES

English: Native

German: A1 – Pursuing

ADDITIONAL INFORMATION

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- Network Intern, Airport Authority of India
 - Hadoop Intern, Ducat Technologies